

Cointegration Analysis for Black Industrial Chains: Hot-Rolled Coil, Iron Ore, Coking Coal, and Rebar

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- **Cointegration Background and Data**
- **Structures for Strategies Used**
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Cointegration Background and Data

- Introduction to Cointegration
- Selection of Futures (HC, I, J, RB)
- Data Cleaning and Preparation (ADF and Johansen Tests)

What is cointegration?

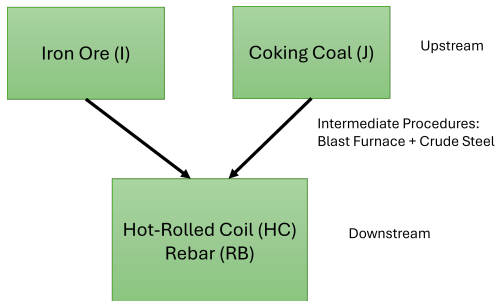
If two or more non-stationary time series, such as asset prices, are cointegrated, there exists a linear combination of them (spread) that is stationary. Although they fluctuate randomly, they do not deviate from the long-term equilibrium and should demonstrate a mean-reverting property.

Processes to develop a cointegration strategy

- Select assets with direct economic or financial connections
- Perform Augmented Dickey–Fuller (ADF) test for stationarity
- Use Johansen test to determine number of relationships and cointegration vector
- Construct spread and verify mean-reversion properties
- Design and backtest a realistic trading strategy using integer contract sizes
- Evaluate performance (Sharpe, win rate, max drawdown, etc.) and improve the strategy

Selection of Futures (HC, I, J, RB)

Logic Behind Cointegrations



Iron ore (I) and coking coal (J) are used in blast furnaces to produce steel, and hot-rolled coil (HC) and rebar (RB) are the final products the process yields. Reduced or increased costs in I or J can cause the same price trend directions in HC and RB prices. The price of all these assets can be simultaneously impacted by macro shocks or global demand of steel in construction or infrastructure. Cointegration is likely for these four contracts because they reflect different stages of the same industrial process. Additionally, China is the world's largest producer and consumer of steel, so domestic prices are globally relevant.

Data Cleaning and Preparation

- Rename columns for clarity (*row*, *date*, *asset*, *open*, *close*).
- Select only necessary columns: *date*, *asset*, *close*.
- Pivot to wide format (each asset as a column).
- Rename asset codes to readable labels: *HC*, *I*, *J*, *RB*.
- Drop rows with missing values.
- Data slicing: use first 833 trading days (2020-07-29 to 2023-12-29) as in-sample, remaining 380 days as out-of-sample (approximately 70% for in-sample data).

Data Cleaning and Preparation (Raw vs. Cleaned)

	A	B	C	D	E
1	number	time	thscode	open	close
2	1	2025-07-29	RB00.SHF	3,204.0000	3,295.0000
3	2	2025-07-28	RB00.SHF	3,260.0000	3,192.0000
4	3	2025-07-25	RB00.SHF	3,217.0000	3,261.0000
5	4	2025-07-24	RB00.SHF	3,223.0000	3,229.0000
6	5	2025-07-23	RB00.SHF	3,228.0000	3,215.0000
7	6	2025-07-22	RB00.SHF	3,135.0000	3,228.0000
8	7	2025-07-21	RB00.SHF	3,077.0000	3,137.0000
9	8	2025-07-18	RB00.SHF	3,071.0000	3,073.0000
10	9	2025-07-17	RB00.SHF	3,051.0000	3,071.0000
11	10	2025-07-16	RB00.SHF	3,045.0000	3,054.0000
12	11	2025-07-15	RB00.SHF	3,115.0000	3,115.0000
13	12	2025-07-14	RB00.SHF	3,100.0000	3,140.0000
14	13	2025-07-11	RB00.SHF	3,127.0000	3,127.0000
15	14	2025-07-10	RB00.SHF	3,084.0000	3,100.0000
16	15	2025-07-09	RB00.SHF	3,055.0000	3,055.0000
17	16	2025-07-08	RB00.SHF	3,070.0000	3,055.0000
18	17	2025-07-07	RB00.SHF	3,090.0000	3,084.0000
19	18	2025-07-04	RB00.SHF	3,073.0000	3,079.0000
20	19	2025-07-03	RB00.SHF	3,041.0000	3,070.0000

	A	B	C	D	E
1	date	HC	I	J	RB
2	2020-07-29	3893	876.5	2135	3840
3	2020-07-30	3921	872	2135	3789
4	2020-07-31	4000	893	2135	3660
5	2020-08-03	4000	900	2135	3780
6	2020-08-04	4030	900	2135	3750
7	2020-08-05	4030	915	2135	3771
8	2020-08-06	4030	910	2135	3771
9	2020-08-07	4030	889	2135	3771
10	2020-08-10	4030	860	2135	3771
11	2020-08-11	4030	863	2135	3760
12	2020-08-12	4030	863	2135	3718
13	2020-08-13	4030	882	2135	3718
14	2020-08-14	4030	920	2135	3718
15	2020-08-17	4030	898.5	1990.5	3718
16	2020-08-18	4068	912	1998.5	3796
17	2020-08-19	4036	915.5	2008.5	3750
18	2020-08-20	3995	914.5	1985	3727
19	2020-08-21	4001	920.5	1980	3729
20	2020-08-24	4007	916.5	1987.5	3752

Data Cleaning and Preparation

ADF Test

- Apply the Augmented Dickey-Fuller (ADF) test to check stationarity in each time series.
- We hope each time series is $I(1)$, namely, integrated of order 1. Alternatively, the time series themselves should be non-stationary, but they should be stationary after first differencing.
- Null hypothesis: the series has a unit root (i.e. non-stationary).
- A low p-value (typically < 0.05) leads to rejection of the null, indicating stationarity.

Data Cleaning and Preparation

ADF Test Results

ADF Test – 0th Difference (I(0))

Asset	p-value
HC	0.457493
I	0.461161
J	0.182159
RB	0.372637

ADF Test – First Difference (I(1))

Asset	p-value
HC	3.43e-11
I	0.000000
J	0.000000
RB	0.000000

Interpretation

All four assets are non-stationary at level ($p > 0.05$), indicating unit roots. However, they become stationary after first differencing ($p < 0.05$), which confirms that they are **I(1)** time series, achieving the basic requirement for cointegration analysis.

Data Cleaning and Preparation

Johansen Test

- The Johansen test identifies the number of cointegration relationships among non-stationary series, which is applicable to our multivariate system involving HC, I, J, and RB.
- Extracted cointegration vectors are used to construct long-run equilibrium relationships (spreads) for trading strategies.

VAR Lag Order Selection (AIC):

Lag	0	1	2	3	4	5	6	7	8	9
AIC	45.41	32.85	32.77	32.78	32.79	32.80	32.82	32.81	32.83	32.84

Interpretation

The minimum AIC occurs at lag 2, making it the optimal lag length for the VAR model before performing the Johansen test. This ensures a balance between model fit and complexity.

Data Cleaning and Preparation

Johansen Test Results

Trace Statistics:

Rank (r)	0	1	2	3
Trace Stat	70.96	33.96	14.72	6.53

Critical Values (90%, 95%, 99%):

Rank (r)	90%	95%	99%
0	44.93	47.85	54.68
1	27.07	29.80	35.47
2	13.43	15.49	19.93
3	2.71	3.84	6.63

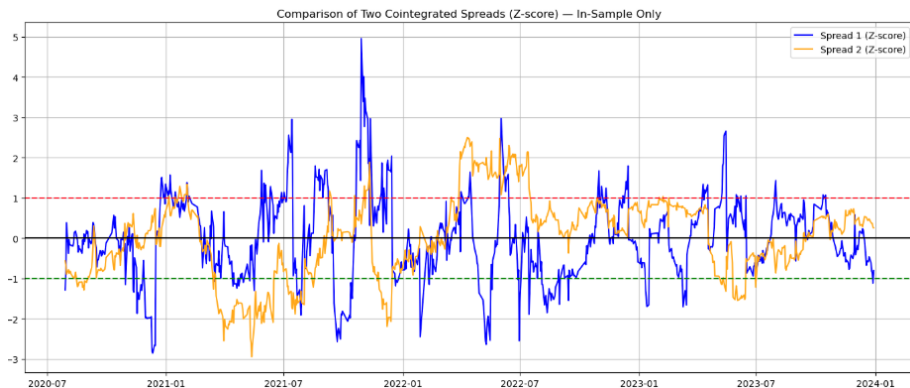
Cointegrating Vectors (Beta):

0.00500051	-0.00229527	-0.00013172	0.00095144
0.0006142	0.0027557	-0.00324887	-0.00065519
0.00156508	0.00212441	-0.00065980	-0.0002007
-0.00657481	0.0007434	-0.0006247	0.0003335

The first two trace statistics exceed their 95% critical values, indicating the presence of **two cointegration relationships** among the four futures. These vectors capture long-run equilibrium, enabling mean-reverting spread construction.

Data Cleaning and Preparation

Spreads of Cointegration Vectors



Spread 1 Equation:

$$S_1(t) = 5.00051 \cdot HC_t + 0.6142 \cdot I_t + 1.56603 \cdot J_t - 6.57481 \cdot RB_t$$

Spread 2 Equation:

$$S_2(t) = -2.29527 \cdot HC_t + 2.7557 \cdot I_t + 2.12441 \cdot J_t + 0.75434 \cdot RB_t$$

Mean-Reversion Assessment (ADF Test):

The Augmented Dickey-Fuller (ADF) p-value for Spread 1 is 0.000000, while Spread 2 has a p-value of 0.042926. Since lower ADF p-values provide stronger evidence against the null hypothesis of a unit root, Spread 1 is more mean-reverting and thus a stronger candidate for statistical arbitrage.

Structures for Strategies Used

- Conversion from Cointegration Vectors to Realistic Hands of Contracts Required for Trading
- Strategy 1: Z-Score (Inspired by Probability Interval Strategy)
- Strategy 2: Plain Vanilla
- Strategy 3: Realized Increment (Exclude for Now)

Structures for Strategies Used

Conversion from Cointegration Vectors to Realistic Hands of Contracts

1. Raw Cointegration Vector

Cointegration Vector = $\beta = \{\text{HC} : 5.00051, \text{I} : 0.6142, \text{J} : 1.56603, \text{RB} : -6.57481\}$

These coefficients reflect the long-term equilibrium relationship. It suggests if we go long, we should theoretically buy 5.00051 tons of HC, 0.6142 tons of I, 1.56603 tons of J, and sell 6.57481 tons of RB, but these are not directly tradable units.

2. Adjust by Contract Sizes

Each commodity futures contract has a different lot size. Specifically, **HC, RB** are 10 tons per contract and **I, J** are 100 tons per contract. We can divide each β_i by its contract size to compute how many full contracts (hands) to buy or sell in real markets. Hence, we have to buy 0.500051, 0.006142, 0.0156603, and -0.657481 hands of HC, I, J, and RB, respectively.

3. Scale and Round to Closest Integers

Multiply the adjusted hands by a relatively large number and round to the closest integer to keep the ratio of coefficients similar to the original cointegration vector to preserve the mean-reverting property. For example, if we scale by 1000, we should buy 500.051 hands of HC, 6.142 hands of I, 15.6603 hands of J, and sell 657.481 hands of RB. After rounding, the number of hands to buy are 500, 6, 16, and -657, respectively. Then, we will take this as the cointegration vector for spread.

Strategy 1: Z-Score

Inspiration

Strategy 2: probability interval (ProbI). If the investor prefers avoiding too much trading in order to save on transaction costs, she should open a position only when the spread deviates significantly from zero. One common choice to quantify this deviation is to trade when the spread exceeds the so-called Bollinger band (see, e.g., [Endres and Stübinger, 2019]), which coincides with the 95% probability interval under a normality assumption, where the two moments of the distribution are estimated dynamically over a rolling window of n days. This improves on the related strategy analyzed in the seminal work of [Gatev et al., 2006], where the bands are just assumed to be constant and equal to twice the sample standard deviation. The present ProbI strategy generalizes the strategy analyzed in [Bock and Mestel, 2009], who consider two different bands associated to two alternating market regimes.

Formally, assuming there is no open position at $t - 1$, a position is opened at time t if $S_t \notin (\hat{\mu}_{t-1-n:t-1} \pm q_\alpha \hat{\sigma}_{t-1-n:t-1})$, where $\hat{\mu}_{n_1:n_2}$ (resp. $\hat{\sigma}_{n_1:n_2}$) is the sample mean (resp. standard deviation) estimated using the datapoints from n_1 to n_2 and q_α is the α -quantile of the standard normal distribution. In our empirical analysis we use the 0.975th quantile. This strategy is equivalent to the one based on the z-score (considered for instance in [Avellaneda and Lee, 2010]), defined as $z_t := \frac{S_t - \hat{\mu}_{t-1-n:t-1}}{\hat{\sigma}_{t-1-n:t-1}}$, that prescribes to open a position at t if $z_t \notin (-q_\alpha, +q_\alpha)$.

Strategy 1: Z-Score

Logic

- **Step 1: Load Futures Data** Aligned daily prices for HC, I, J, and RB are imported from 2020–2025.
- **Step 2: Scale Cointegration Vector** Original cointegration coefficients (from Johansen test) are scaled by 1000 and adjusted for contract size to produce realistic integer "hands" (contracts) for trading.
- **Step 3: Construct Spread with Integer β** A new spread series is computed using:

$$\text{Spread}_t = \beta^\top P_t = \sum_i \text{Contracts}_i \times \text{Price}_i$$

- **Step 4: Rolling Z-Score Calculation** Mean and standard deviation over a 31-day rolling window define:

$$z_t = \frac{\text{Spread}_t - \mu_t}{\sigma_t}$$

- **Step 5: Signal Generation**
 - Go Long if $z_t < -1.2$
 - Go Short if $z_t > 1.2$
 - Exit if $z_t \cdot z_{t-1} < 0$ (crosses mean)
- **Step 6: Position Tracking** Positions are updated daily and held until exit. The position can be 1 (long), -1 (short), or 0 (flat).

Strategy 1: Z-Score

Logic

- **Step 7: PnL and Exposure Calculation** For each trading day:
 - **Spread Change PnL:** $\Delta \text{Spread}_t \times \text{Position}_{t-1}$
 - **Transaction Costs:** Subtract 0.5% of gross notional if the position changes
 - **Exposure:**
 - Gross Exposure = Sum of |notional positions|
 - Net Exposure = |sum of notional positions|
- **Step 8: Save Results to Excel** Full position history, PnL, and notional exposure per asset saved for backtest review.
- **Step 9: Key Performance Metrics**
 - Final cumulative profit = cumulative PnL after all trades
- **Step 10: Visualization**
 - Spread and Z-score bands with thresholds plotted
 - Cumulative profit curve over time
- **Goal:** Evaluate how realistic integer-scaled trading performs under real contract sizes, incorporating transaction costs and practical signal generation.

Strategy 2: Plain Vanilla Mean-Reversion

Inspiration

Strategy 1: plain vanilla (PV). This first strategy is a benchmark in the pairs trading literature (see [Burgess, 1999]). According to this rule, the investor should open a position as soon as the spread differs from zero. This strategy implies that the investor exploits every deviation of the assets from their long-run relationship and might be highly profitable. However, this rule involves very frequent trading (since the spread oscillates frequently around zero) and, as shown below, might not be profitable as soon as transaction costs are taken into account.

Formally, assuming there is no open position at $t - 1$, a position is opened at t if $S_t \neq 0$.

Strategy 2: Plain Vanilla Mean-Reversion

Logic

- **Data Preparation:** Use full aligned futures data for HC, I, J, and RB. Construct spread using scaled and rounded integer-valued cointegration vector (scaled by 1000 and converted to contract units).

- **Spread Construction:**

$$\text{Spread}_t = \sum_i \text{Contracts}_i \times \text{Price}_{i,t}$$

- **Rolling Mean (130-day):**

$$\mu_t = \frac{1}{130} \sum_{k=t-129}^t \text{Spread}_k \Rightarrow \text{Demeaned}_t = \text{Spread}_t - \mu_t$$

- **Entry/Exit Logic:**

- Enter **Long** when spread is below mean $\rightarrow \text{spread} - \text{mean} < 0$
- Enter **Short** when spread is above mean $\rightarrow \text{spread} - \text{mean} > 0$
- **Exit** when spread crosses back over the mean $\rightarrow$ sign changes

- **Position Tracking:** Positions are held until the spread mean is crossed. Position is updated daily.

Strategy 2: Plain Vanilla Mean-Reversion

Logic

- **PnL Calculation:**

- Profit from spread change:

$$\text{PnL}_t = (\text{Spread}_t - \text{Spread}_{t-1}) \times \text{Position}_{t-1}$$

- Subtract transaction cost (0.5%) when position changes:

$$\text{Cost} = 0.005 \times \text{Gross Exposure}$$

- **Exposure Calculation:**

- **Gross Exposure** = sum of absolute notional values across assets
- **Net Exposure** = absolute sum of signed notional values

- **Performance Metrics:**

- Final Cumulative Profit: cumulative PnL after all trades

- **Visualization:**

- Cumulative Profit over time
- Spread with rolling mean overlaid

Judging Parameters and Correlation

- Parameter Definitions
- Scoring Methods for Composite Scores
- Heatmaps and Graphs relating parameters and composite scores

Parameter Definitions for Composite Score

- **Sharpe Ratio (S):** Measures risk-adjusted return.

$$S = \frac{\mu_r}{\sigma_r} \sqrt{252}$$

where μ_r is the average daily return, and σ_r is the standard deviation of daily returns.

- **Maximum Drawdown (D):** Measures the worst peak-to-trough decline in cumulative profit.

$$D = \max_t \left(\max_{s \leq t} P_s - P_t \right)$$

where P_t is the cumulative profit at time t .

- **Win Rate (W):** Proportion of profitable trading days.

$$W = \frac{\text{Number of Profitable Days}}{\text{Number of Trading Days with Position}}$$

Parameter Definitions for Composite Score (cont'd)

- **Number of Trades (T):** Total number of position entries. More trades can imply better opportunity capture.
- **Average Holding Time (H):** Mean duration (in days) a trade is held. Shorter holding times are preferred for liquidity.

Composite Score (C) is computed as (three weightings):

$$C = 0.20 \cdot S_N + 0.10 \cdot D_N + 0.10 \cdot W_N + 0.35 \cdot T_N + 0.25 \cdot H_N$$

$$C = 0.25 \cdot S_N + 0.15 \cdot D_N + 0.10 \cdot W_N + 0.25 \cdot T_N + 0.25 \cdot H_N$$

$$C = 0.30 \cdot S_N + 0.20 \cdot D_N + 0.15 \cdot W_N + 0.20 \cdot T_N + 0.15 \cdot H_N$$

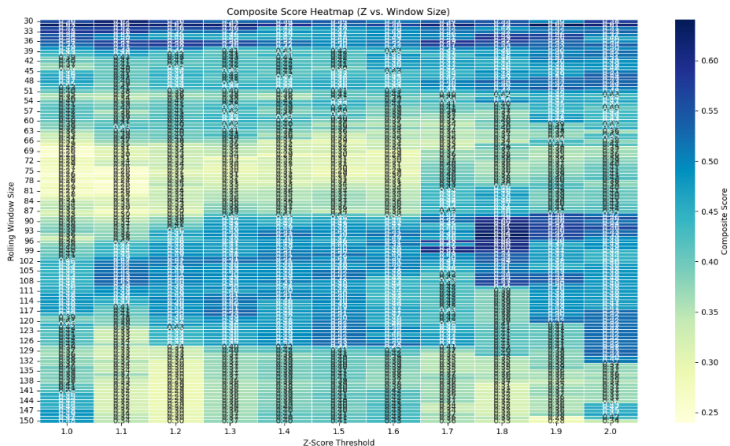
where each metric is normalized to $[0, 1]$ range: S_N , D_N , W_N , T_N , and H_N .

Effect of Parameter Weight on Z-Score and Window

- **↑ Sharpe Ratio Weight:** Favors high return with low volatility. $\Rightarrow$ Optimizer prefers **moderate z-score** and **moderate-to-long window** for smooth and balanced signals.
- **↑ Max Drawdown Weight:** Penalizes large losses. $\Rightarrow$ Strategy becomes more conservative with **higher z-score** and **longer window** to reduce reversals.
- **↑ Win Rate Weight:** Rewards consistent success. $\Rightarrow$ Optimizer favors **higher z-score** and **moderate window** for more accurate entries.
- **↑ Trade Count Weight:** Encourages frequent trading opportunities. $\Rightarrow$ Optimizer shifts toward **lower z-score** and **shorter window** to generate more signals.
- **↑ Weight on Short Holding Time:** Promotes quick mean-reversion execution. $\Rightarrow$ Strategy favors **lower z-score** and **shorter window** to exit positions faster.

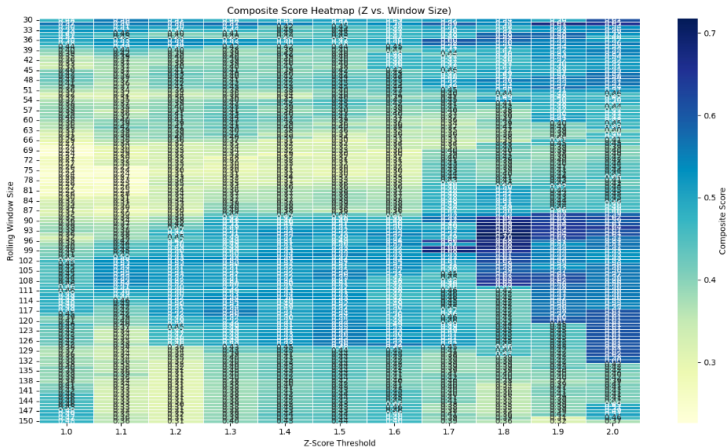
Heatmap 1 for Z-Score Strategy (40% Parameters Boosting Z-Scores)

$$C = 0.20 \cdot S_N + 0.10 \cdot D_N + 0.10 \cdot W_N + 0.35 \cdot T_N + 0.25 \cdot H_N$$



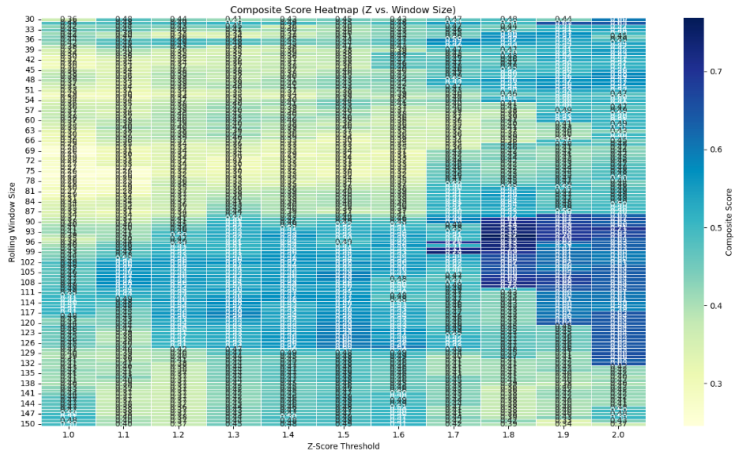
Heatmap 2 for Z-Score Strategy (50% Parameters Boosting Z-Scores)

$$C = 0.25 \cdot S_N + 0.15 \cdot D_N + 0.10 \cdot W_N + 0.25 \cdot T_N + 0.25 \cdot H_N$$



Heatmap 3 for Z-Score Strategy (65% Parameters Boosting Z-Scores)

$$C = 0.30 \cdot S_N + 0.20 \cdot D_N + 0.15 \cdot W_N + 0.20 \cdot T_N + 0.15 \cdot H_N$$



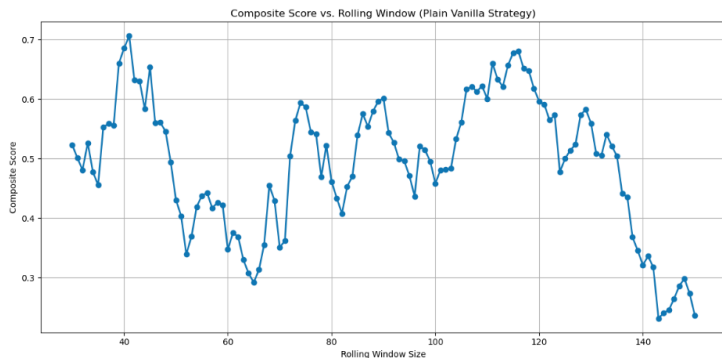
Graph 1 for Plain Vanilla Strategy (60% Parameters Reducing Window Sizes)

$$C = 0.20 \cdot S_N + 0.10 \cdot D_N + 0.10 \cdot W_N + 0.35 \cdot T_N + 0.25 \cdot H_N$$



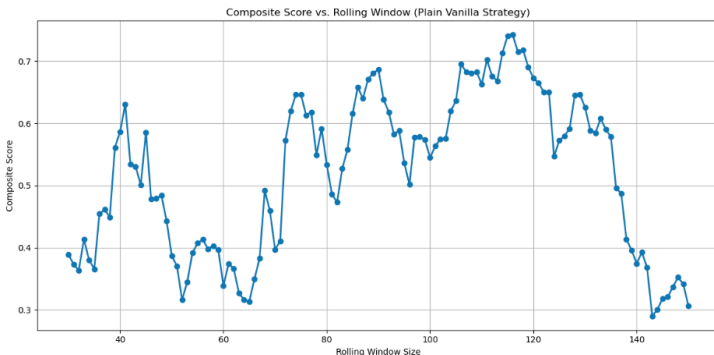
Graph 2 for Plain Vanilla Strategy (50% Parameters Reducing Window Sizes)

$$C = 0.25 \cdot S_N + 0.15 \cdot D_N + 0.10 \cdot W_N + 0.25 \cdot T_N + 0.25 \cdot H_N$$



Graph 3 for Plain Vanilla Strategy (35% Parameters Reducing Window Sizes)

$$C = 0.30 \cdot S_N + 0.20 \cdot D_N + 0.15 \cdot W_N + 0.20 \cdot T_N + 0.15 \cdot H_N$$



Results

- Best Three Strategies from Composite Score Analysis
- Optimal Strategy based on Cumulative PnL on Test Data
- Graph of the Optimal Strategy

Results

Best Three Strategies

Summary of Optimal Parameters

- **Z-Score Strategy:** All three weighting schemes indicate the best performance occurs at **z-score ± 1.8** and **window size 94**.
- **Plain Vanilla Strategy:**
 - First two weightings: Optimal at **window size 41**
 - Third weighting: Optimal at **window size 116**

Goal

Compare the performance of these three strategies by using test data from **2024-01-01** to **2025-07-29** to determine which strategy generates the highest profit.

Results

Optimal Strategy based on Cumulative PnL on Test Data

Profit Comparison (Test Period: 2024-01-01 to 2025-07-29)

- Z-Score Strategy ($z = \pm 1.8$, window = 94):

$$41769690 - 38766090 = 3,003,600$$

- Plain Vanilla Strategy (window = 41):

$$28781351.65 - 25892832.4 = 2,888,519.25$$

- Plain Vanilla Strategy (window = 116):

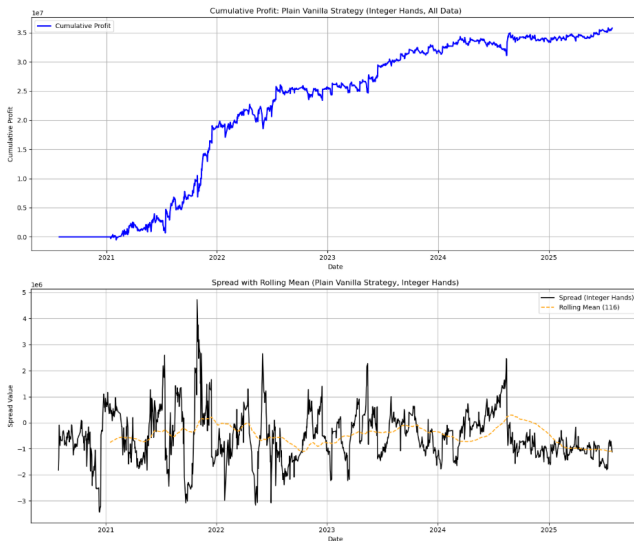
$$35791343.95 - 31836586.9 = 3,954,757.05$$

Conclusion

Among the three top strategies, the **Plain Vanilla Strategy with window size 116** yields the highest profit on the test data, making it the best-performing strategy during this period.

Results

Graph of the Optimal Strategy



Market Events and Volatility

- Major Profit Drawdowns
- Event Correlation with Spread
- Volatility Analysis

Possible Future Topics

- Regime-Switching Spread Models
- Incorporating Macro Factors
- Deep Learning for Prediction